

OpenClaw AI Diet Calorie Estimation Tutorial

Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on the diet calorie estimation project itself.

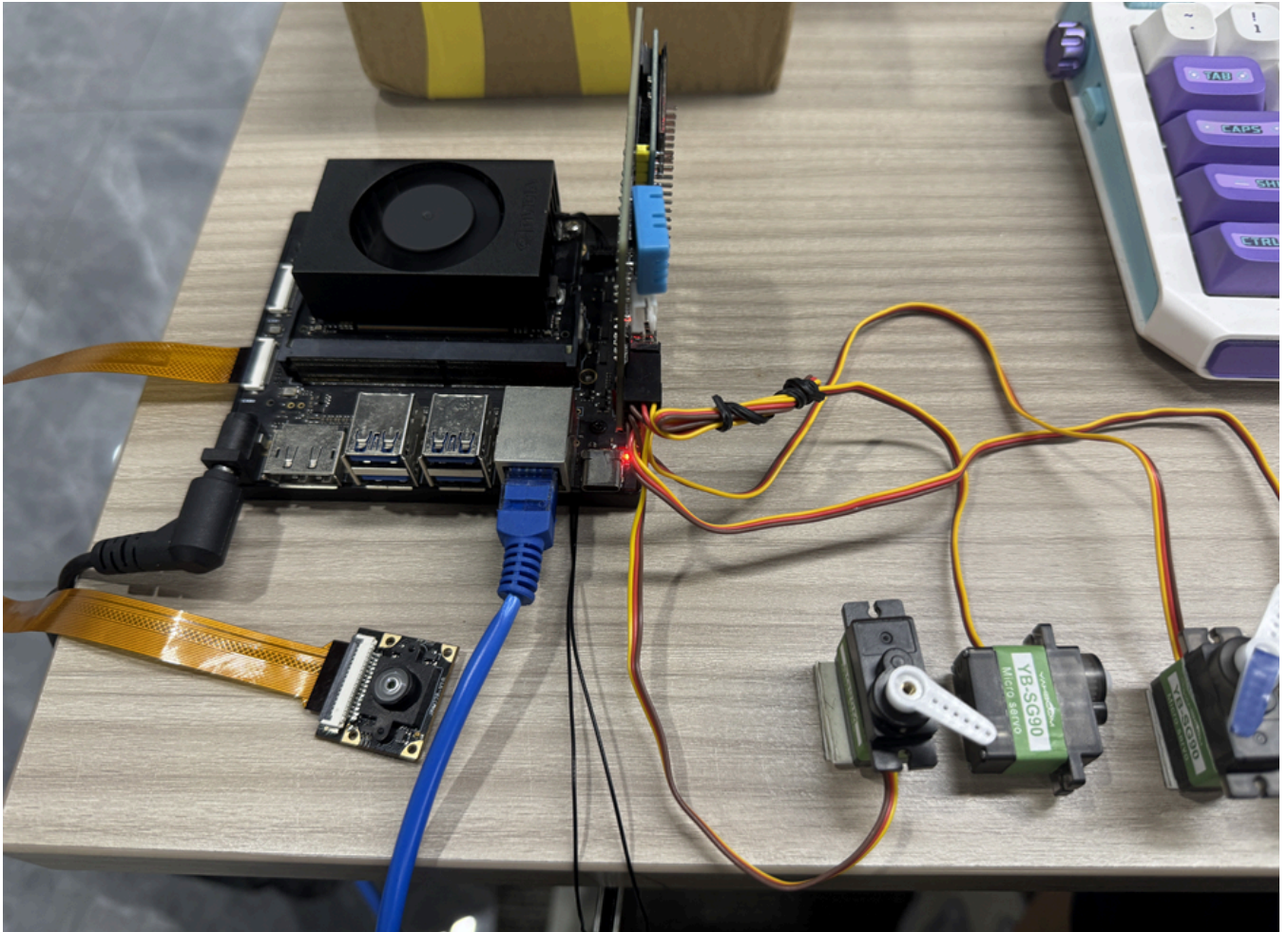
1. Tutorial Objectives

This article demonstrates OpenClaw's natural language capabilities in diet analysis scenarios through "photo recognition + large model estimation".

- Understand how to use cameras and visual models for plate analysis
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Complete four project scenarios: plate photo, food recognition, calorie estimation, health advice

2. Hardware Preparation

- OpenClaw mainboard
- CSI camera module
- OLED display
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points that will be used in the tutorial

- Camera photo command: `/home/jetson/hardware_ha1_demo/camera`
- Photo CLI: `/home/jetson/hardware_ha1_demo/cli/seewhat`
- Image output directory: `/home/jetson/hardware_ha1_demo/cli`
- Voice entry: `/home/jetson/voice_to_openc1aw/src/main.py`

In this project, the real "calorie estimation" is mainly completed by large models, so it is recommended to ensure the following two pipelines are normal first:

- Camera can complete photo capture
- Visual model can complete image description

Note: Calorie estimation is more suitable as a demonstration and daily reference, not suitable as medical-grade or nutritionist-level precise measurement results.

4. Test Project Basic Capabilities

Before starting this project, please complete one `openc1aw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After passing, enter TUI to verify the basic capabilities in the project.

First enter OpenClaw TUI:

```
openclaw tui
```

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
connected | idle
agent main | session main (openclaw-tui) | baillian/qwen3.5-plus | tokens 0/1.0m (0%)
```

Then you can first do the most basic photo and recognition test, for example:

- Help me take a lunch photo
- Look at what's on this plate

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote diet recording, classroom demonstrations	Convenient for remote viewing results after taking photos	Demonstrate "visual + dialogue" capability
TUI	Local debugging, model joint debugging	Most direct, easiest to verify analysis results	First run through photo and description chain
Voice code	Voice Q&A, life assistant experience	Closer to daily usage method	As complete demonstration solution

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 whatsapp Solution

For whatsapp connection and robot creation steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. After creation, you can remotely initiate diet analysis in whatsapp, for example:

- Take a photo to see current food, estimate the food calories. If it exceeds 1000 calories, light up red light, otherwise light up green light. Servo rotates to 90 degrees to indicate not recommended for consumption, 0 degrees indicates recommended for consumption

20:23



< 2



+86 153 3885 7526 (你)

给自己发消息

Take a photo to see the current food, estimate the food calories. If it exceeds 1000 calories, light up red light; otherwise light up green light. servo rotates to 90 degrees to indicate not recommended for consumption, 0 degrees indicates recommended for consumption.

20:22 ✓✓



I can see a **Coca-Cola can** in the image. Let me estimate the calories and provide the appropriate feedback.

Calorie Estimate: A standard 330ml Coca-Cola can contains approximately **140-150 calories**.



Since this is well under 1000 calories



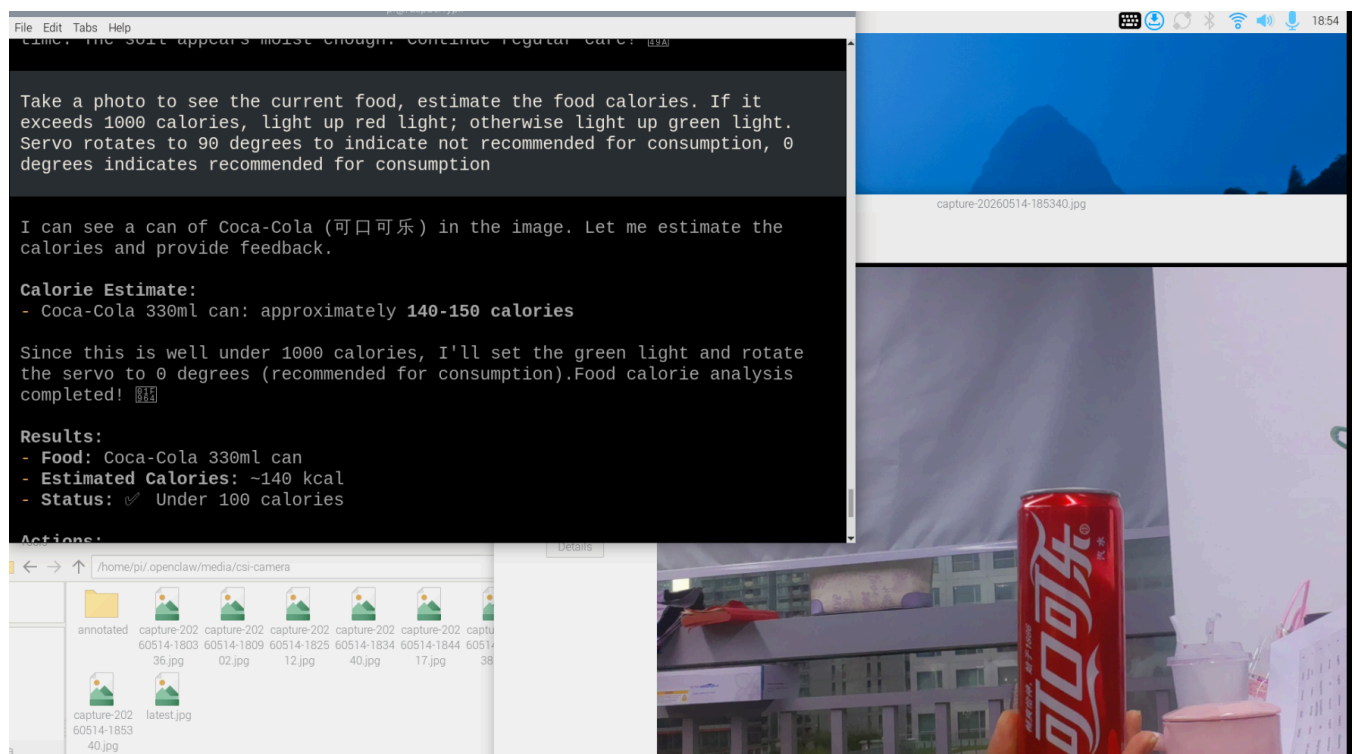
5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then input project-related commands, for example:

- Take a photo to see current food, estimate the food calories. If it exceeds 1000 calories, light up red light, otherwise light up green light. Servo rotates to 90 degrees to indicate not recommended for consumption, 0 degrees indicates recommended for consumption



5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After waking up, you can directly say control content, for example:

- Food calorie recognition

```
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 3750 ms 语音，正在转写
```

```
[你] Food, calorie recognition.
```

```
[发送] 正在交给 OpenClaw
```

```
[完成] OpenClaw 返回 200
```

```
[OpenClaw] ## 📷 Food Calorie Recognition Complete!
```

```
**Identified:** Coca-Cola
                (可口可乐) - 330ml can
```

```
**Nutritional Information:**
```

```
- **Calories:**
```

```
    ~140 kcal
```

```
- **Sugar:** ~35g
```

```
- **Carbohydrates:** ~37g
```

```
-
```

```
    **Caffeine:** ~32mg
```

```
**Health Note:** This is a regular (non-diet)
```

```
    Coca-Cola. One can contains about 7% of the recommended daily
    calorie intake for an average adult, but nearly all of it comes from
    sugar. 📷
```

```
The calorie information is now displayed on the OLED
```

```
screen! Would you like me to recognize any other food items? 📷
```

```
[信息] 回复已裁剪为播报文本：511 -> 121 字
```

```
- [播报] 正在播放回复
```

6. Project Implementation

Note: The following are all dialogue demonstrations using `openclaw tui` method. You can also choose whatsapp or voice method to complete.

6.1 Scenario One: Plate Photo

Experimental Objective

First complete plate image capture to get an image available for subsequent analysis.

Effect Description

After successful photo capture, a plate image will be saved locally and generate a most recent image alias. Subsequent recognition and estimation will continue based on this image.

Example Instructions

- Help me take a lunch photo
- Take a plate photo now

6.2 Scenario Two: Food Recognition

Experimental Objective

Let OpenClaw recognize what main foods are on the plate.

Effect Description

The focus of this step is not immediately giving calories, but first recognizing main contents like rice, meat, vegetables, drinks, etc., laying the foundation for subsequent estimation.

Example Instructions

- Look at what foods are in this meal
- Help me recognize the plate contents
- List the main ingredients

6.3 Scenario Three: Calorie Estimation

Experimental Objective

Based on recognized food content, let OpenClaw give an approximate calorie estimation.

Effect Description

OpenClaw can give total calories or itemized calories based on image content and common food experience values. This result is more suitable for demonstration and reference, not suitable as strict nutritional calculation.

Example Instructions

- Estimate the total calories of this meal
- Estimate itemized by rice, eggs, meat, and vegetables
- How many calories is this lunch approximately

6.4 Scenario Four: Health Advice

Experimental Objective

Let OpenClaw give a brief suggestion after estimating calories.

Effect Description

Besides reporting numbers, OpenClaw can also give a suggestion closer to daily communication based on goals like "fat loss, muscle gain, light meal, late-night snack".

Example Instructions

- If I'm losing weight, give me a suggestion
- Is this meal combination balanced
- Help me summarize the result into one sentence

Extensible Gameplay

- Display total calories on OLED
- Make into daily diet check-in assistant
- Combine with voice broadcast to form life scenario demonstration